

CSC1003 Practice 3

Conditionals, Loops and Arrays

Screenshot

- Please do not take a picture of the computer screen with your phone when asking questions in the WeChat group. Instead, take a direct screenshot.
- The shortcuts:
Windows: Windows + Shift + S
Mac: Command + Shift + 5(3/4)

Recommended Reading: How To Ask Questions The Smart Way

English Version: <http://www.catb.org/~esr/faqs/smart-questions.html>

Chinese Version: https://github.com/ryanhanwu/How-To-Ask-Questions-The-Smart-Way/blob/main/README-zh_CN.md

Scanner

- OJ only support Scanner for the input.
- A demo:

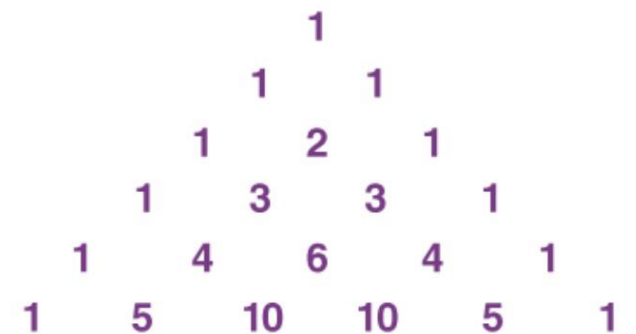
```
import java.util.*;  
public class Main {  
    Run | Debug  
    public static void main(String[] args) {  
        Scanner scanner = new Scanner(System.in);  
        int n, m;  
        n = scanner.nextInt();  
        m = scanner.nextInt();  
        scanner.close();  
    }  
}
```

Divisible

- Write a program that takes two positive integers and prints true if either evenly divides the other. You might need % (to check divisible) and boolean operations.
- OJ link: https://oj.cuhk.edu.cn/d/csc1003_2025_fall/p/P1005

Pascal's Triangle

- Pascal's Triangle is the triangular arrangement of numbers that gives the coefficients in the expansion of any binomial expression. The numbers are so arranged that they reflect as a triangle. Firstly, 1 is placed at the top, and then we start putting the numbers in a triangular pattern. The numbers which we get in each step are the addition of the above two numbers. It is similar to the concept of triangular numbers. Below is a draft of the triangle.
- Given an integer n ($1 \leq n \leq 12$), write a program to output the first n lines of the pascal triangle.
- OJ link: https://oj.cuhk.edu.cn/d/csc1003_2025_fall/p/P1006



Reverse

- Read n integers and output the in reverse order.
- Hint: Consider `.split()` method.
- Java String documentation:
<https://docs.oracle.com/javase/8/docs/api/java/lang/String.html>
- OJ link: https://oj.cuhk.edu.cn/d/csc1003_2025_fall/p/P1007

Divisor

- Write a Java program to compute the greatest common divisor of two given positive integers x and y . For example, the highest common factor of 54 and 24 is 6.
- OJ link: https://oj.cuhk.edu.cn/d/csc1003_2025_fall/p/P1008

Q&A